CHE 321 Chemical Reaction Engineering

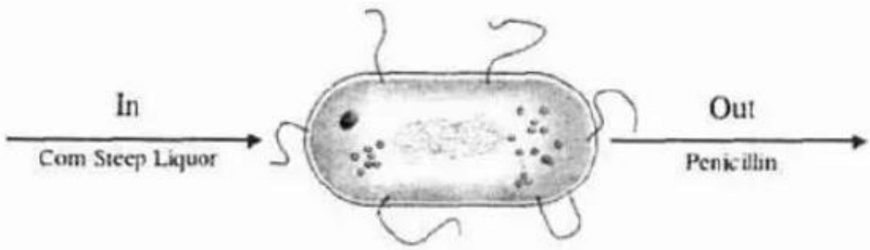
Homework #1 (Due: September 5th midnight)

(please submit solutions as one combined pdf)

1. Textbook (H Scott, Fogler. 5th edition), P1-4B.

2.

P1-11$_B$ We are going to consider the cell as a reactor. The nutrient corn steep liquor enters the cell of the microorganism *Penicillium chrysogenum* and is decomposed to form such products as amino acids, RNA, and DNA. Write an unsteady mass balance on (a) the corn steep liquor, (b) RNA, and (c) pencillin. Assume the cell is well mixed and that RNA remains inside the cell.



Penicillium chrysogenum

NOTE: P1-4B need to use the Smog in Los Angeles Basin Web Module. (http://umich.edu/~elements/5e/web_mod/la_basin/index.htm)

Use MATLAB to solve ODE, below is the link and instructions

For MATALB, you can access it free through UIC web store: https://webstore.illinois.edu/shop/product.aspx?zpid=3819

and there is some instructions:
https://www.mathworks.com/help/matlab/math/choose-an-ode-solver.html
https://www.mathworks.com/help/matlab/ref/ode45.html

P1-4B

700 squaremiles $(2\times10^{10} ft^2$

inversion height $= 2000$ ft

Volume of air $4\times10^{13} ft^3$

(a)

Given

$T = 75°F \Rightarrow 534.67\ R$

$P = 1 atm$

$V = 4\times10^{13} ft^3$

desired unit

lb * mol

use ideal gas law

$PV = nRT$

$$n = \frac{PV}{RT}$$

$$R = 0.7302 \frac{(ft^3)(atm)}{(lb)(mol)(R)}$$

$$n = \frac{(1\cancel{atm})(4\times10^{13}\cancel{ft^3})}{\left(0.7302 \frac{(\cancel{ft^3})(\cancel{atm})}{(lb)(mol)(\cancel{R})}\right)(534.67\cancel{R})}$$

$$n = \frac{4\times10^{13}}{390.42(lb)(mol)}$$

$$n = 1.025\times10^{11}(lb)(mol)$$

b) Given

$P = 1\,atm$

$V = 3000\ \frac{ft^3}{h}$

$T = 32°F \Rightarrow 491.67\ R$

$R = 0.7302$

$CO = 2\% \Rightarrow 0.02$

$PV = nRT$

$$n = \frac{PV}{RT} \Rightarrow \frac{(\cancel{1\,atm})(3000\ \cancel{ft^3}/h)}{\left(0.7302\ \frac{\cancel{(ft^3)}\cancel{(atm)}}{(lb)(mol)\cancel{(R)}}\right)(491.67\ \cancel{R})}$$

$$n = \left(8.35614\,01645\ \frac{(lb)(mol)}{(h)}\right)(400{,}000)$$

$$= 3{,}342{,}456.0658\ \frac{(lb)(mol)}{h}\ (0.02\ CO)$$

$$= 66849.12\ \frac{(lb)(mol\ CO)}{h}$$

c) Given

$V = 15\ mph \Rightarrow 79200\ ft/h$

$$A = (2000\,ft)(20\,\cancel{mi})(5280\ ft/\cancel{mi}) = 211{,}200{,}000\,ft^2$$

$$Q = (79200\ ft/h)(211{,}200{,}000\ ft^2)$$

$$Q = 1.67 \times 10^{13}\ ft^3/hr$$

d) $In - Out + \cancel{Gen}^{0} - \cancel{Con}^{0} = Acc$

$In = F_{CO,A} + F_{CO,S}$

$Out = (V_0)(C_{CO})$

$Acc = \frac{dN_{CO}}{dt} \Rightarrow \frac{d(VC_{CO})}{dt} \Rightarrow (V)\frac{(d \cdot CCO)}{dt}$

$\therefore$

$$F_{CO,A} + F_{CO,S} - V_0 C_{CO} = (V)\frac{(d \cdot CCO)}{dt}$$

e)

$$F_{CO,A} + F_{CO,S} - V_0 C_{CO} = (V)\frac{(d \cdot CCO)}{dt}$$

$$dt = \frac{V\,dC_{CO}}{[F_{CO,A} + F_{CO,S}] - V_0 \times C_{CO}}$$

$$\frac{dt}{V} = \frac{dx}{(a - bx)}$$

$$\int_0^t \frac{dt}{V} = \int_{C_{CO,0}}^{C_{CO,f}} \frac{dx}{a - bx}$$

$$\frac{V_0 t}{V} = -\ln([F_{CO,A} + F_{CO,S}] - V_0 C_{CO,f}) + \ln([F_{CO,A} + F_{CO,S}] - V_0 C_{CO,0})$$

$$\frac{V_0 t}{V} = \ln \frac{[F_{CO,A} + F_{CO,S}]}{[F_{CO,A} + F_{CO,S}] - V_0 (C_{COf})}$$

$$t = \frac{V}{V_0} \ln \left(\frac{[F_{CO,A} + F_{CO,S}]}{[F_{CO,A} + F_{CO,S}] - V_0 (C_{COf})} \right)$$

f. given

$V = 4 \times 10^{13} ft^3$

$V_0 = V_{cars} + V_{wind} \Rightarrow \left(3{,}000 \frac{ft^3}{h}\right)(400{,}000) + 1.67 \times 10^{13} \frac{ft^3}{h}$

$V_0 = 1.6728 \times 10^{13} \frac{ft^3}{h}$

$F_{CO,A} = 6.68 \times 10^4 \frac{lb(\text{moles of CO})}{h}$

$C_{CO,0} = 2.04 \times 10^{-8} \frac{lb(\text{mol of CO})}{ft^3}$

$C_{CO,F} = 2 \text{ ppm} \times \frac{1}{1{,}000{,}000 \text{ lb mol}} \times \frac{1.024 \times 10^{11} \text{ lb mol}}{4 \times 10^{13} ft^3} \Rightarrow 5.12 \times 10^{-9} \frac{lb(\text{mol CO})}{ft^3}$

$n = 1.024 \times 10^{11} \text{ lb (mol of air)}$

$$t = \frac{V}{V_0} \ln \left(\frac{[F_{CO,A} + F_{CO,S}] - V_0 C_{CO,0}}{[F_{CO,A} + F_{CO,S}] - V_0 C_{COf}} \right)$$

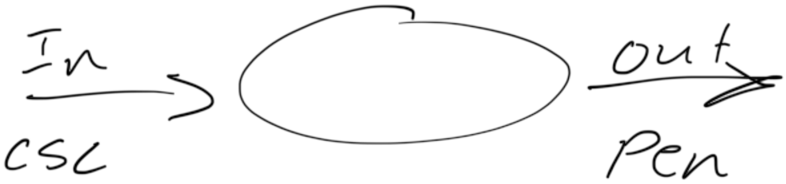
$F_{CO,A} + F_{CO,S} = 6.68 \times 10^4 \frac{lb(\text{mol CO})}{h} + 3412.13 \frac{lb(\text{mol CO})}{h}$

$$t = \frac{4 \times 10^{13} ft^3}{1.6728 \times 10^{13} \frac{ft^3}{hr}} \ln \left(\frac{[F_{CO,A} + F_{CO,S}] - (1.6728 \times 10^{13} \frac{ft^3}{h})(2.04 \times 10^{-8} \frac{lb(\text{mol CO})}{ft^3})}{[F_{CO,A} + F_{CO,S}] - (1.6728 \times 10^{13} \frac{ft^3}{h})(5.12 \times 10^{-9} \frac{lb(\text{mol CO})}{ft^3})} \right)$$

$t = 6.85 \text{ hr}$

Over time the 10 concentration goes down with slight spikes but the overall trend is that the graph is trending negatively.

Q 2)



$$\text{in} - \text{out} + \text{gen} - \text{con} = \frac{dN_{CSL}}{dt}$$

a) CSL goes in Pen comes out

$$F_{CSL_0} - C_{CSL} = \frac{dN_{CSL}}{dt}$$

b) NO RNA in, NO RNA out, but RNA is present so...

$$\frac{dN_{RNA}}{dt} = G_{RNA}$$

Generation

c) No Pen in only Gen & out

$$\frac{dN_{pen}}{dt} = G_{pen} - F_{pen}$$

out